Reg. No. | | | | | | | | | | |

[14CS/IS46]

FOURTH SEMESTER B.TECH UNIVERSITY EXAMINATION, JUNE – 2017

DESIGN AND ANALYSIS OF ALGORITHMS

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Instructions: Answer any FIVE full questions choosing at least TWO questions from each Part.
2. Any missing data may be suitably assumed

PART – A

1. a. What is an algorithm? Illustrate the important properties of an algorithm, with example. **08**

b. Discuss the 3 methods of calculating GCD with suitable example. **06**

2. a. Explain best case, worst case and average case analysis. **03**

b. If T1(n)=O(f(n)) and T2(n)= O(g(n)), then show that T1(n) + T2(n) = O{max(f(n),g(n))}. **06**

c. Consider the following algorithm, **05**

```
ALGORITHM ENIGMA(A[0...n-1,0....n-1])
for i← 0 to n-2 do
  for j←i+1 to n-1 do
    if A[i,j]≠A[j,i]
      return FALSE
end for
end for
  return TRUE
```

Determine the following:

i. What does the algorithm compute?

ii. What is its basic operation?

iii. How many times is the basic operation executed?

iv. Analyze the efficiency class of this algorithm?

v. Can this algorithm be further improved?

3. a. What is Brute force method? Explain with an example, how would this approach is applied to a job assignment problem? **07**

b. Explain with an example, linear search algorithm. Find its best, worst and average case efficiency. **07**

4. a. Explain the divide and conquer methodology. Suggest a pseudocode for merge sort and analyze its complexities. Trace algorithm to the data set 8,4,1,6,7,2,3,9. **08**

b. Trace the quicksort algorithm to sort the list "QUESTION" in alphabetical order. Draw the tree of recursive calls made. **06**

PART - B

5. a. Differentiate between depth first search and breadth first search technique. **05**

b. Define the following with example. **04**

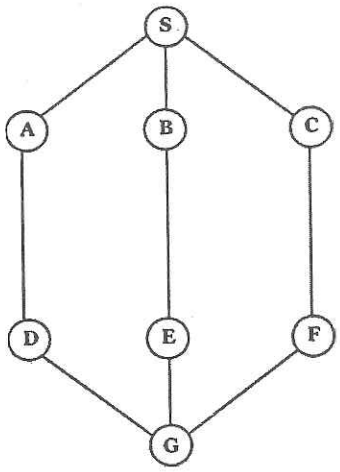
i. Directed graph

ii. Weighted graph

iii. Connected graph

iv. Adjacency matrix

c. Apply DFS algorithm to find DFS order for the vertices of the following graph. Write algorithm for the same. **05**

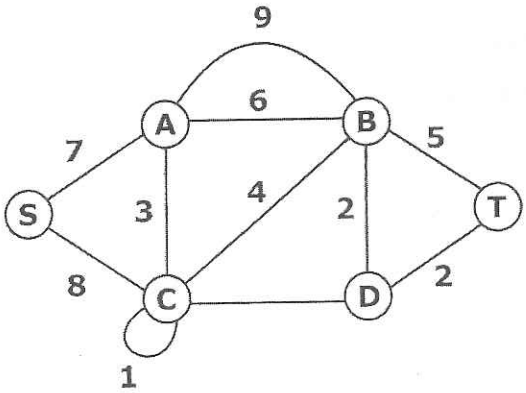


6. a. Apply Warshall's algorithm to find the transitive closure of the graph defined by the following adjacency matrix. Write Warshall's algorithm. **07**

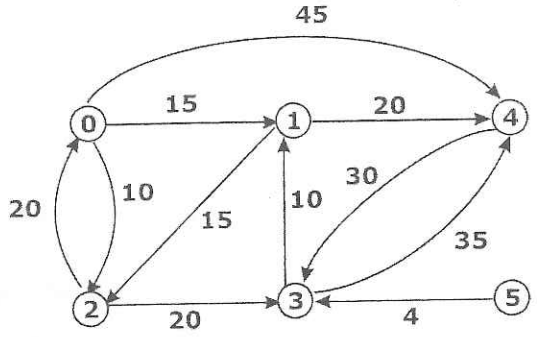
$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

b. Explain with an example, how to compute binomial coefficient. Derive its time complexity. **07**

7. a. Apply Prim's algorithm to find minimum spanning tree for the graph shown below. **07**



b. Using Dijkstra's algorithm, trace the following graph to get shortest path from vertex 5 to all other vertices. **07**



8. **a.** Apply the Horspoo's pattern matching algorithm for the text "THIS-IS-A-SIMPLE-EXAMPLE" and match pattern EXAMPLE. **10**

b. Define the following: **04**

i. Encryption

ii. Decryption

iii. Plain text

iv. Cipher text